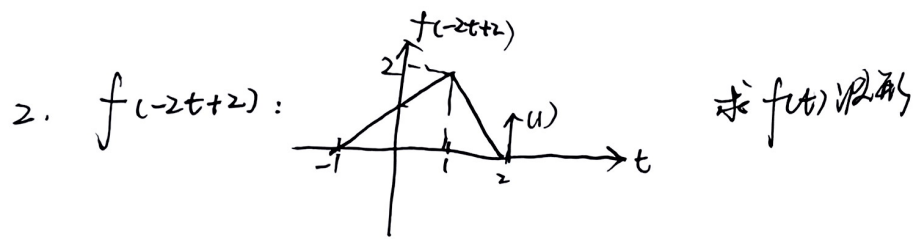


1. $\frac{dr(t)}{dt} + t r(t) + \int_{-\infty}^t r(x) dx = \frac{de(t)}{dt} + 3e(t)$ 是否线性时不变。

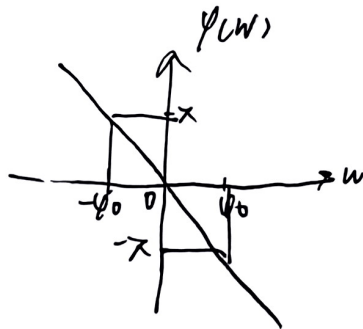
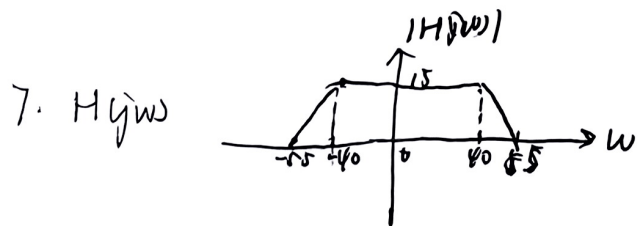


3. $f_1(t) = 3[\varepsilon(t-2) - \varepsilon(t-5)]$, $f_2(t) = \varepsilon(t-1) - \varepsilon(t+1)$ 求 $f(t) = f_1(t) * f_2(t)$ 画出波形

4. $\frac{d^2 r(t)}{dt^2} + 4 \frac{dr(t)}{dt} + 4r(t) = \frac{de(t)}{dt} + 3e(t)$; $r(0)=1, r'(0)=2$ 求 $r(t)$ 与 $h(t)$

5. 傅里叶尺度变换特性叙述与证明

6. $F(s) = \frac{s^2}{s^2 + 3s + 2} e^{-s}$, 求 $f(t)$



$e(t) = 1 + \cos(2t) \cdot \cos(\varphi t) + \sin(3t) + \cos(w)$

求 $f(t)$, 判是否因果系统 + 理由



7. ~~求~~ $\omega_0 = 20$

8. $f(t) = e^{-2|t|}$, $t \in (-\infty, +\infty)$ 求双边 Laplace 及 42 分 E

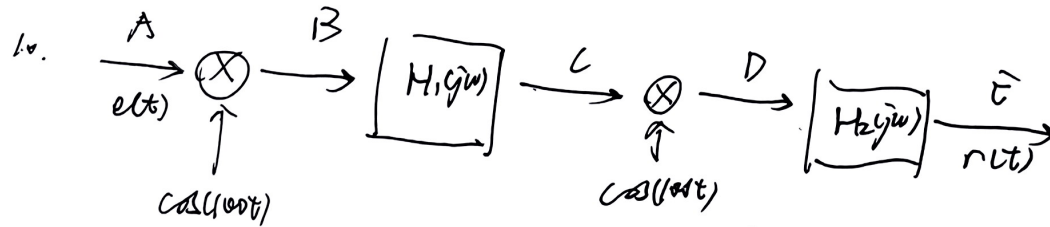
9. $e(t) = 30 \cos(20\pi t) + 50 \sin(40\pi t)$

$a(t) = [100 + e(t)] \cos(2000\pi t)$ V

1). $a(t)$ 频谱图, 求频宽

2). $R = 50 \Omega$ 求 $\bar{P}$, P_s , P_c

3). 是否可用包络检波法解调, 理由



$e(t) = 50 \pi t / \pi t$, $H_1(j\omega)$ $H_2(j\omega)$ $\varphi(\omega) \approx$

1). $e(t)$ PT

2). A, B, C, D 频谱

3). 求 $r(t)$

4). 是否失真, 理由

